



MATH ANSWER KEY – 25 QUESTIONS

Test: Comprehensive Mid-Term
★ Total: 100 pts (Base) + 20 Bonus

Date: June 17, 2026
⌚ Time: 90 min

Contents

1. Section A: Multiple Choice (Q1–10) – 40 pts
2. Section B: Algebra (Q11–15) – 25 pts
3. Section C: Geometry (Q16–20) – 25 pts
4. Section D: Mixed / Word Problems (Q21–25) – 10 pts
5. Bonus (Q26) – 20 pts extra
6. Grading Rubric

Section A: Multiple Choice (40 pts, 4 pts each)

#	Answer	Explanation & Common Mistake
1	B) 5	$3x + 7 = 22 \rightarrow 3x = 15 \rightarrow x = 5$. ⚠ Don't forget to subtract 7 first.
2	C) 10	$2(x-3)=14 \rightarrow x-3=7 \rightarrow x=10$. ⚠ Distribute before dividing.
3	A) 25π	Area = $\pi r^2 = \pi(5)^2 = 25\pi$. ⚠ Don't use circumference formula.
4	D) 7	Data: 2,3,3,5,7,7,7,9 $\rightarrow$ 7 appears most (3 times).
5	B) $3x^2 - 10x - 8$	$(3x+2)(x-4)=3x^2-12x+2x-8=3x^2-10x-8$.
6	A) $3x + 10$	$5x+3-2x+7 = (5x-2x)+(3+7)=3x+10$.
7	C) 7	$4y-5=2y+9 \rightarrow 2y=14 \rightarrow y=7$. ⚠ Watch signs when moving terms.
8	B) 3	Slope = $(8-2)/(3-1) = 6/2 = 3$.
9	D) 13	$5^2+12^2 = 25+144=169 \rightarrow \sqrt{169}=13$. ⚠ Hypotenuse is always the longest side.
10	A) 64	Volume of cube = $s^3 = 4^3 = 64$.

Section B: Algebra – Show Work (25 pts)

Q11 (5 pts) Solve: $5x - 2 = 3x + 8$

1. $5x - 3x = 8 + 2 \rightarrow 2x = 10$

2. $x = 5$

✓ $x = 5$

2 pts setup, 3 pts answer

Q12 (5 pts) Solve: $3(y+2) = 2(y-1)$

1. $3y+6 = 2y-2 \rightarrow 3y-2y = -2-6$

2. $y = -8$

✓ $y = -8$

2 pts distribution, 3 pts solving

Q13 (5 pts) System: $x+y=8$, $x-y=2$

1. Add: $2x = 10 \rightarrow x = 5$

2. Substitute: $5+y=8 \rightarrow y=3$

✓ $(5, 3)$

3 pts x, 2 pts y

Q14 (5 pts) Expand: $(2x-3)^2$

1. $(2x)^2 - 2(2x)(3) + 3^2$

2. $4x^2 - 12x + 9$

✓ $4x^2 - 12x + 9$

3 pts expansion, 2 pts combining

Q15 (5 pts) Factor: $x^2 - 5x + 6$

1. Find factors of 6 that sum to -5: -2 and -3

2. $(x-2)(x-3)$

✓ $(x-2)(x-3)$

2 pts factors, 3 pts correct binomials

Section C: Geometry – Show Work (25 pts)

Q16 (5 pts) Area of triangle: base=10, height=6

$$A = \frac{1}{2} \times b \times h = \frac{1}{2} \times 10 \times 6 = \mathbf{30} \text{ sq units}$$

2 pts formula, 3 pts computation

Q17 (5 pts) Perimeter of rectangle: L=8, W=5

$$P = 2(L+W) = 2(8+5) = 2(13) = \mathbf{26} \text{ units}$$

2 pts formula, 3 pts answer

Q18 (5 pts) Circumference of circle: r=7

$$C = 2\pi r = 2\pi(7) = \mathbf{14\pi} \text{ units}$$

2 pts formula, 3 pts correct

Q19 (5 pts) Hypotenuse: legs 9 and 12

$$1. c^2 = 9^2 + 12^2 = 81 + 144 = 225$$

$$2. c = \sqrt{225} = \mathbf{15}$$

2 pts setup, 3 pts answer

Q20 (5 pts) Volume of cylinder: r=3, h=5

$$V = \pi r^2 h = \pi(3^2)(5) = \pi(9)(5) = \mathbf{45\pi} \text{ cubic units}$$

2 pts formula, 3 pts computation

Section D: Mixed / Word Problems (10 pts, 2 pts each)

Q21 Simplify: $(x^2)(x^3)$

$\mathbf{x^5}$

(add exponents)

Q22 Solve: $x/3 + 4 = 7$

$\mathbf{x = 9}$

(subtract 4, multiply by 3)

Q23 Mean of: 2,4,6,8,10

$\mathbf{6}$

(sum=30, divide by 5)

Q24 Slope of $y = 2x + 3$

$\mathbf{2}$

(coefficient of x)

Q25 40% as a fraction

$\mathbf{2/5}$

(40/100 = 2/5)

★ Bonus (20 pts extra)

Q26 (Challenge) Solve: $\sqrt{3x + 1} = x - 1$

1. Square both sides: $3x + 1 = (x-1)^2$
2. Expand: $3x + 1 = x^2 - 2x + 1$
3. Set to zero: $0 = x^2 - 5x \rightarrow x(x-5)=0$
4. Solutions: $x=0$ or $x=5$. Check for extraneous: $x=0$ gives $\sqrt{1} = -1$? False. $x=5$ gives $\sqrt{16} = 4$, and $5-1=4$ ✓

✓ **Final Answer: $x = 5$**

5 pts squaring, 5 expansion, 5 factoring, 5 extraneous check

📋 Grading Rubric (Teacher Use)

Section A (MCQ) Q1–10: $10 \times 4 = 40$
pts

Section B (Algebra) Q11–15: $5 \times 5 = 25$
pts

Section C (Geometry) Q16–20: $5 \times 5 = 25$
pts

Section D (Mixed) Q21–25: $5 \times 2 = 10$
pts

✦ **Base total:** $40+25+25+10 = 100$ pts | **Bonus:** Q26 up to 20 pts

💡 Partial Credit Guidelines

- Algebra: 50% for correct setup, 100% for final correct.
- Geometry: full points for correct formula use even if arithmetic is off by 1.
- Bonus: only full credit if extraneous solution is checked and rejected.